ISSN 0335 - 3931

NF F 50-010
December 1993

French standard

Classification index: **F 50-010**

Railway fixed equipment

Cam bolts - Oval head bolts

French standard approved by decision of the Director General of AFNOR on 20 November 1993 to take effect on 20 December 1993.

Replaces the approved standard with the same index dated July 1990.

Correspondence

As of the date of publication of the present standard, there is no draft standard or European or international standard covering the same subject.

Analysis

The present standard is part of a series of documents defining certain accessories intended to be used to fix rails on concrete sleepers.

Describers

International Technical Thesaurus: railway fixed equipment, rail track, rail, sleeper, reinforced concrete fixing element, bolt, manufacture, characteristic, dimension, testing, marking, description.

Modifications

Compared with the previous issue:
- details of dimensional tolerances
- modification of mechanical characteristics
- updated to comply with standards in force.

Corrections

Published by the Bureau de Normalisation des Chemins de Fer (BNCF), 15 rue Traversière, 75571 PARIS CEDEX 12, Tel.: (1) 40 19 15 03
Circulated by BNCF and AFNOR - Tour Europe, 92049 Paris La Défense, Tel.: (1) 42 91 55 55

Photocomposition MIT
First issue 94-02

Contents

1. Scope

The present standard covers the supply of cam bolts (figure 1) used to fix track rails on concrete sleepers. This standard also covers the supply of oval head bolts (figure 2) used to indirectly fix rails and track equipment on sleepers and concrete supports.

EXAMPLES

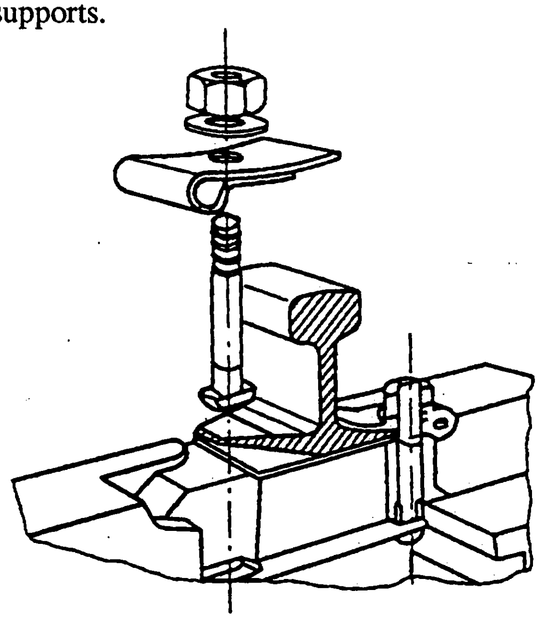


Figure 1: Cam bolt

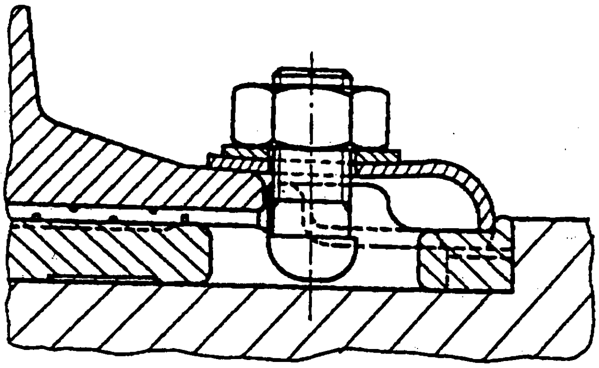


Figure 2: Oval head bolt

The cam bolts and oval head bolts are hereinafter referred to as "bolts".

2. References

This French standard includes by dated or undated reference the stipulations of other publications. These standard references are given at the appropriate place in the text and are listed below. For dated references, subsequent amendments or revisions to any one of these publications only apply to this French standard insofar as they have been incorporated by amendment or revision. For the undated references, the most recent issue of the publication to which references are made shall apply.

* *Translator's note: the following may not exactly correspond to the official AFNOR translation.*

NF EN 10204	Metal products - Types of inspection documents (classification index: A 00-001)
NF A 91-102	Metal coatings - Electro-plated zinc and cadmium coatings on iron or steel
NF A 91-111	Metal coatings - Thickness measurement - Coulometric method using anodic dissolution
NF A 91-112	Metal coatings - Thickness measurement using the magnetic method - General specifications
NF A 91-113	Metal coatings - (Surface treatment) - Measurement of thicknesses using Foucault currents - General specifications
NF A 91-121	Zinc plating by immersion in molten zinc - (Hot-dip galvanization) - Finished products in iron - Steel - Cast iron

NF A 91 - 122	Metal coatings - Finished products in hot-dip zinc-plated steel - Recommendations relative to the design and use of zinc plated products.
NF A 91-472	Chromating of zinc or cadmium plating - Specifications and tests methods
NF E 03-001	Triangular metric threads - ISO profile
NF E 03-053	Triangular metric threads - ISO system of thread tolerances - Average quality 6 H/6 g for bolts and other current applications.
NF E 03-151	Triangular metric threads - Verification of threads using limit gauges - General
NF E 03-152	Triangular metric threads - Verification of threads using limit gauges - Threaded gauges for external threads
NF E 03-154	Triangular metric threads - Verification of threads using limit gauges - Smooth gauges for screws and taps
NF E 25-007	Fasteners - Conditions for ordering and delivery
NF E 25-009	Fasteners - Electrolytic coatings on threaded components
NF E 25-019	Fasteners - Ends of ISO external threaded parts
NF EN 20898-1	Mechanical characteristics of fixing parts - Part 1: Bolts, screws and pins (Classification index: E 25-100-1)
NF EN 24014	Fixing parts - Partly threaded hex. head screws - Grades A and B (Classification index: E25-112)
NF E 27-024	Bolt hardware - Tolerances for threaded bolts
NF F 00-004	General railway equipment - Cast, matrixed or machine engraved characters
NF F 00-800	General railway equipment - Definition of procedures for certification of products - General rules
NF F 50-007	Railway fixed equipment - Large hex. nuts, grade B - Nut HL
NF EN 29002	Quality systems - Model for production and installation quality assurance (Classification index: X 50-132).

3. Descriptions

Cam bolts or oval head bolts which comply with the present standard are hereinafter designated to by:
- the term "cam bolt" or "oval head bolt";
- the diameter;
- the length under the head;
- the threaded length;
- the reference of the present standard;
- the nut designation.

EXAMPLE: Cam bolt 22 - 150/50, NF F 50-010, nut HL 22.

4. Fabrication

4.1 Materials

The steel grades used must enable the mechanical characteristics of finished parts as described in Section 5.3 to be obtained.

4.2 Fabrication method

The bolt shanks are fabricated as single parts without welding and the heads by hot upsetting.
The nuts are as forged and executed as per the stipulations of standard NF F 50-007.

4.3 Threads

The threads are executed to ISO profile as per standard NF E 03-001, to quality 6 H/6 g as per standard NF E 03-053.

The thread is executed by cold rolling for fabrication of the electro-plated zinc shank (this process is not required for the fabrication of bolt shanks of a diameter equal to or above 20 m, black or galvanized)

4.4 Corrosion protection

The bolts can be delivered untreated or with electrolytic or hot-dip zinc plating.

The anti-corrosion protection is applied as per the specifications of the following standards:

- NF A 91-121 and A 91-122 for hot-dip galvanization
- NF A 91-102, NF A 91-472 and NF E 25-009 for zinc electro-plating.

When hot-dip galvanization is applied, the minimum quantity of zinc must be type Z6 as per standard NF A 55-101 and the weight of the deposited zinc must be equal or greater than 200 g/m2.

In the case of zinc electro-plating, the protection must be type Zn 12/D/Fe.

Other protections can be used. The order must stipulate their nature and their thickness.

5. Characteristics

5.1 Physical characteristics

The bolts must be clean, uniform and present no defects (rolling folds, cutting or thread burrs, missing material or local overthicknesses).

The shafts must be smooth, free from scale or cracks. The heads must show no folding of the metal at their junction with the shaft.

The bearing faces of the heads must be free from any roughness resulting from deburring which could be prejudicial to tightening.

5.2 Dimensional characteristics

5.2.1 Dimensions

The dimensions of the bolts are given by:

- figure 3 for the 18 cam bolt;
- figure 4 for the 22 cam bolt;
- figure 5 for the 20 and 22 oval head bolts.

All the other dimensions can be specified with the order or on its annexed documents.

The length of the shaft under the head and the threaded lengths are stipulated in standard NF EN 24014.

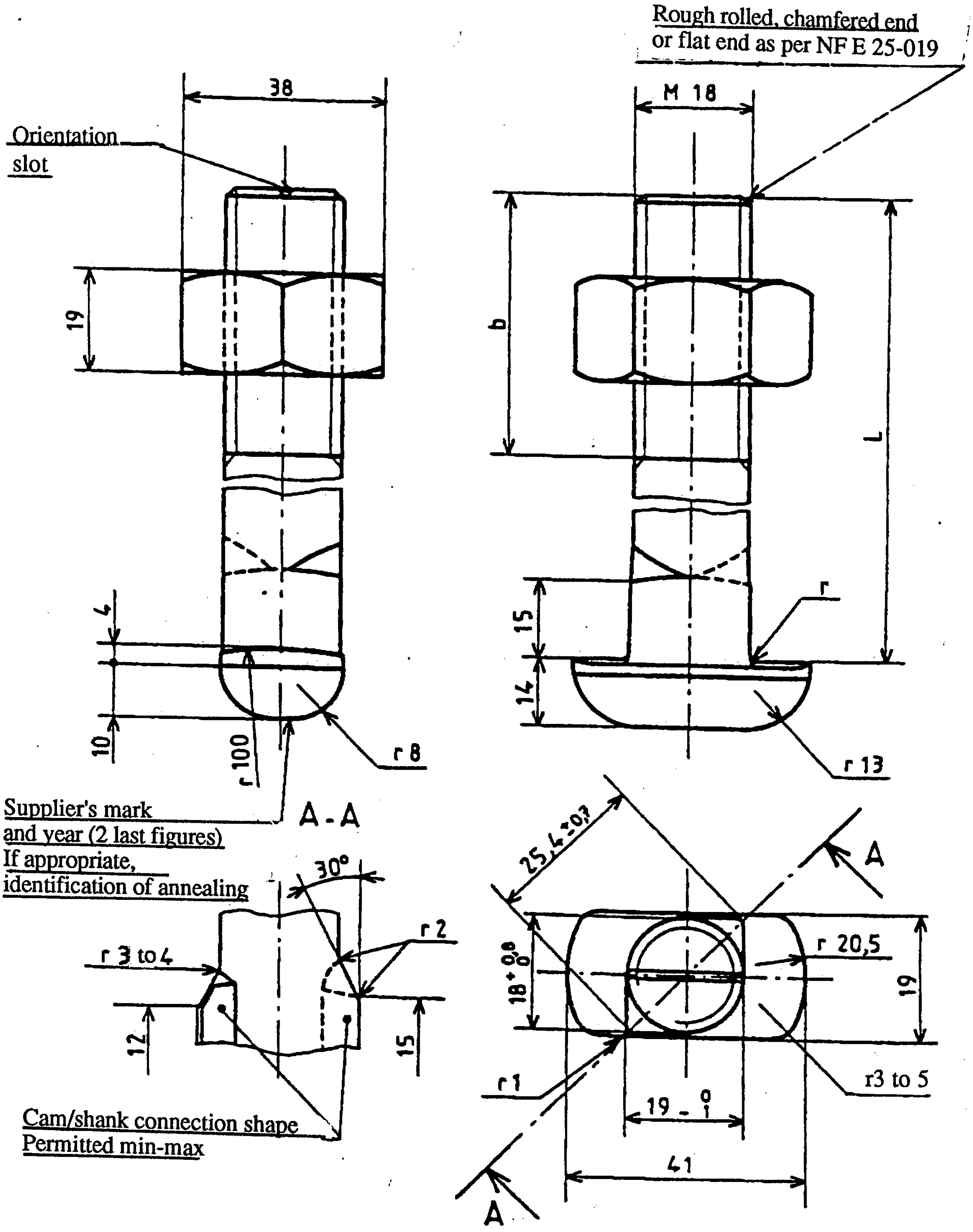


Figure 3: M 18 cam bolt

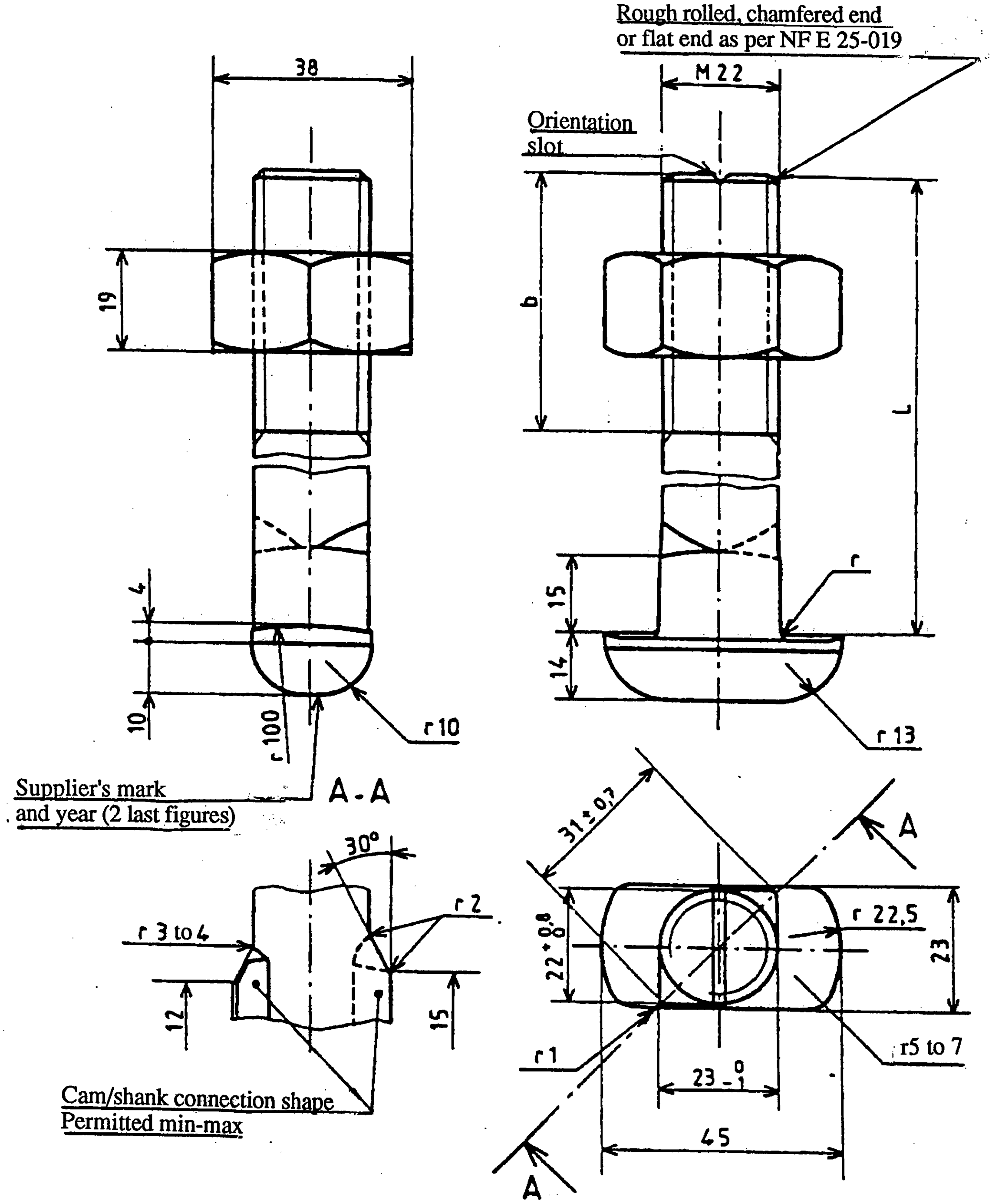


Figure 4: M 22 cam bolt

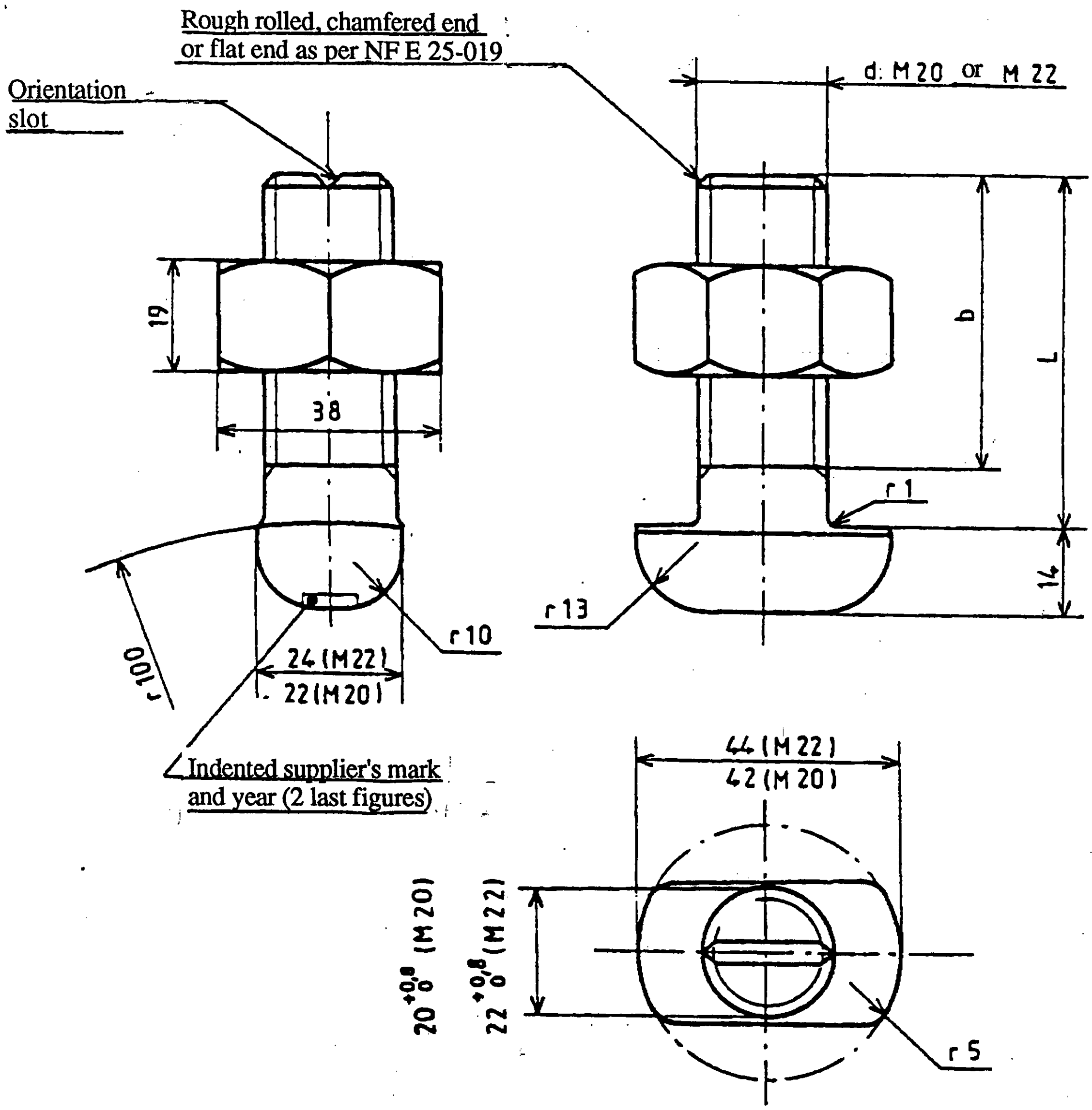


Figure 5: Oval head bolt

5.2.2 - Tolerances

The tolerances to be complied with when manufacturing bolts are specified in figures 3, 4 and 5, the order or its associated documents. These tolerances are determined based on the general standard for bolts NF E 27-024, apart from those which apply to threaded parts which are as follows:

- for the threads on bolt shanks delivered without coating, the tolerance class is 6 g,
- for the threads on bolts delivered zinc electro-plated, the discrepancy before application of a protective coating is 6 thk - 50 µm and the class of tolerance after protection is 6 g,
- the class of tolerance for threads prior to galvanization is 6 g,
- for the nut, the inner thread is quality 6 H or 6 AZ as per the specification of standard NF F 50-007.

These classes of tolerance are defined in standard NF E 03-053.

5.3 Mechanical characteristics

- For the 18 mm diameter bolt shank:
  - the yield strength must be equal to or greater than 600 N/mm2,
  - elongation (A%), for a length Lo equal to 5d, must be equal to or greater than 17%,
  - the minimum individual failure energy (KU) at 20°C must be equal to or greater than 20 J.

- For bolt shanks of 20 and 22 mm diameters:
  - the failure strength must be equal to or greater than 500 N/mm2,
  - elongation (A%), for Lo equal to 5d, must be equal to or greater than 17%,
  - the minimum individual value for the failure energy (KU) at 20°C must be equal to or greater than 25 J.

Tensile strength tests and impact bending tests are executed as per the stipulations of standard NF EN 20890-1.
The folding test of the bolt shank is carried out as per section 7.3. During this test no cracking or break must occur.

Nuts:

The nuts must be executed in class 5 as per standard NF F 50-007.

6. Sampling

The nature of the tests, the acceptable level of quality (AQL) and the level of inspection are given in table 1.
The sampling plan is given in Annex A.

Table 1

Test	AQL	Nature(1)	Level of inspection
Inspection of appearance (folding, missing metal, etc.)	2.5	0	S3
Dimensional characteristics and shape	2.5	0	S3
- Length under head	2.5	0	S3
- Threaded length	2.5	0	S3
- Cam height	2.5	0	S3
- Head thickness	2.5	0	S3
- Head length	2.5	0	S3
- Head width	2.5	0	S3
- Slot orientation	2.5	0	S3
- Threaded ring ENTERS 6g 2)	2.5	0	S4
- Threaded ring DOES NOT ENTER 6g 2)	2.5	0	S4
Mechanical characteristics			
- Tensile strength test of entire part	4	0	S3
- Impact bending test	4	0	S1
- Folding test	4	0	S3
Quality of protective lining	1.5 3)	0	S3

1) 0 is the systematic inspection of characteristics
2) In the case of bolt bodies delivered "untreated" or zinc electro-plated or before galvanization
3) In the case of hot-dip galvanization, the AQL is 4.

7. Test methods and results

7.1 Tensile strength test

The bolt is attached at either end to a test machine. The assembly is executed so as to guarantee perfect centring of the part to undergo testing.

Failure must never occur at the connection of the shaft with the head. The failure load reduced to the initial section must be greater than the minimum value set out for the material used in section 5.3.

7.2 Impact bending test

The impact bending test must be carried out in accordance with the stipulation of standard NF EN 20898-1. The minimum individual value to be obtained must be the one given for the material in Section 5.3.

7.3 Folding test

The bolt shank is folded, if possible in its smooth part, to an angle of 30°.
If the length of the smooth part does not allow the folding test to be executed, this test is carried out on a blank with the same conditions of delivery as the finished parts.
The shank of the bolt to be tested is inserted at least up to half its length in a hole in a cast iron or steel block. The diameter of the hole must not be greater than 1.1 times the diameter of the bolt shank. The hole opening must be rounded down to a radius less than or equal to half the nominal diameter of the bolt shank.

7.4 Inspection of the anti-corrosion protection

The inspection of the quality of the anti-corrosion protection is carried out in accordance with the stipulations of the order and its associated documents and in compliance with, standards NF A 91-102, NF A 91-111, NF A 91-112, NF A 91-113, NF A 91-121, A 91-122, NF A 91-472 and NF E 25-009, as appropriate.

7.5 Dimensional inspection

The dimensional inspection is carried out on sampled bolt shanks drawn from the fabrication batches.
The external diameter, the threaded length, the length under the head and the dimensions of the head are checked using gauges and callipers.
The qualities 6H, 6AZ and 6g of the inner and outer threads are inspected using min-max. threaded master gauges and rings as per the stipulations of standards NF E 03-151, NF E 03-152 and NF E 03-154.

7.6 Results

Any non-conforming results will lead to the corresponding batch being rejected.
However, and only when the dimensional characteristics or the appearance are inadequate, the supplier has the right to renew presentation of the incriminated batch after a unit sort has been performed.

8. Marking and identification

The shank of the bolt has the following marks struck in relief or indent (see figures 3, 4 and 5) as per the stipulations of standard NF F 00-004 with a minimum 4mm height of characters:

- the supplier's mark,
- the two last figures of the year of manufacture,
- for 18 dia. cam bolts, a distinctive identification for restoration annealing.

9. Inspections

9.1 Inspection of the material

The inspection of the raw material used does not have to be submitted to the quality service of the customer railway company, and the supplier has the right to supply material covered by a specific inspection carried out in the production plant. However, delivery must be accompanied by a plant inspection document for the products as described in Section 3.1.B or 3.2 of standard NF EN 10204.

9.2 Batching

The products are presented grouped in batches. Each batch must only include parts of the same type and of the same dimensions manufactured in the same conditions.

9.3 Nature and proportion of tests

The bolts are submitted to inspections and tests using statistical sampling as per table 1 or any other method agreed by the client.

9.4 Presentation

The bolts are normally presented for acceptance in their condition at delivery, i.e., after application of any anti-corrosion treatment.

However, it is accepted that they may be presented before undergoing the protective treatment when this is carried out in another establishment than that of the supplier, holder of the order. This latter nevertheless retains entire responsibility for the quality of the supplies delivered. He is then entitled to take all necessary steps to have the quality of the protective treatment checked.
The date of availability for presentation to the quality department of the customer railway company is stipulated in a written memo issued by the management of the production plant and its authorized representative.
This memo must indicate the number of parts in each batch together with the references of the order which covers these parts.
During presentation, a certificate confirming compliance with the conditions of fabrication and inspection and stipulating the results is handed over to the customer's inspector of fabrications.
Series tests may be carried out unless otherwise stipulated in the order or in any other associated documents, without the presence of the representative of customer railway company. In all cases, the results of the tests are communicated or made available to the latter.

10. Quality assurance

10.1 Model for quality assurance

The model for standard NF EN 29002 quality assurance can be specified in documents prepared when awarding the contract.

10.2 Control and supervision of the quality of productions

The supplier must implement an organization, the methods and means to enable him to check the quality and to monitor his outputs together with any other elements making up the supplies, and prepare a quality plan which implements these stipulations.
The customer railway company checks by specific actions of audit, supervision and inspection, exercised in the supplier's establishments, that the stipulations implemented are effective.
The methods of work adopted by the customer railway company in relation to its own regulations are stipulated in the order.

11. Certification

The track bolts must satisfy the certification procedure imposed by the manager of the infrastructure in compliance with the specifications of standard NF F 00-800.

12. Traceability

The supplier must prepare and keep operational a system for identifying and tracing the constituent products and elements at all stages of production, inspection and delivery.

He must check the traceability of operations, products and elements constituting the products that he sub-contracts in order to be able to easily retrace the history of any data concerning them, interrelate between them and to attribute them with certitude to affected products.
Traceability will concern the following points in particular:

- the origin of the material,
- method of preparation and contractual characteristics,
- operations for transformation and treatment,
- dimensional inspections.

Traceability must make it possible to check all the parts involved in the same fabrication and inspection operations.

13. Delivery

The order indicates the nature of packaging required for each supply together with the quantity of parts per packaging.
The packaging used for shipping, must not have been used previously for any purpose which may be liable to cause damage to the packaged parts.

Unless otherwise indicated in the order, each packaging carries a securely fixed label legibly and indelibly marked with the following indications as per standard NF E 25-007:

- the name and the make of the supplier,
- the order number,
- the nature and the quantity of products packed,
- the reference of the product standard,
- the type of lining.

NOTE: the supplier is entitled to complete this identification by any information which may be needed for the traceability of the product.

14. Guarantee

The bolts are guaranteed by the supplier in normal conditions of use for a period of two years against any defects which can be attributed to fabrication and not detected on acceptance in the plant.

The bolts which during the period of guarantee are revealed to have fabrication defects making them unsuitable for use or which may shorten their service life are rejected.

The rejected bolts must be replaced at the expense of the supplier within a period of one month following the date of notice sent to him.

Rejected products are kept available to the supplier with a view to replacement.

15. Archiving

Independently of any regulations, the archiving method (documents, delays, precautions, etc.) for data carrying the proof of conformity of the products with the present standard together with the specific conditions of the order or of associated documents are defined by the customer railway company which so informs the supplier. All precautions must be taken to avoid deterioration of the documents during the period of archiving.

ANNEX A (integral part of the standard)

Sampling table - Single plan

Quantity of parts of batch to undergo acceptance								Accepted Rejected	AQL (or NQA)					LQ10 for AQL of :				
Levels of special inspection				Levels of inspection														
S 1	S 2	S 3	S 4	I	II	III			0.65	1	1.5	2.5	4	0.65	1	1.5	2.5	4
2 500	2 150	2 50	2 25	2 50	2 15	2 8	Sampling Nb of defects	A R	20 0 1	13 0 1	8 0 1	5 0 1	3 0 1	11	16	25	37	54
501 35 000	151 1 200	51 150	26 90	51 90	16 25	9 15	Sampling Nb of defects	A R	20 0 1	13 0 1	8 0 1	5 0 1	3 0 1	11	16	25	37	54
35 001 ∞	1 201 35 000	151 500	91 150	91 150	26 50	16 25	Sampling Nb of defects	A R	20 0 1	13 0 1	8 0 1	5 0 1	13 1 2	11	16	25	37	27
	35 001 ∞	501 3 200	151 500	151 280	51 90	26 50	Sampling Nb of defects	A R	20 0 1	13 0 1	8 0 1	20 1 2	13 1 2	11	16	25	18	27
		3 201 35 000	501 1 200	281 500	91 150	51 90	Sampling Nb of defects	A R	20 0 1	13 0 1	32 1 2	20 1 2	20 2 3	11	16	12	18	25
		35 001 500 000	1 201 10 000	501 1 200	151 280	91 150	Sampling Nb of defects	A R	20 0 1	50 1 2	32 1 2	32 2 3	32 3 4	11	7.6	12	16	20
		500 001 ∞	10 001 35 000	1 201 3 200	281 500	151 280	Sampling Nb of defects	A R	80 1 2	50 1 2	50 2 3	50 3 4	50 5 6	4.8	7.6	10	13	18
			35 001 500 000	3 201 10 000	501 1 200	281 500	Sampling Nb of defects	A R	80 1 2	80 2 3	80 3 4	80 5 6	80 7 8	4.8	6.5	8.2	11	14
			500 001 ∞	10 001 35 000	1 201 3 200	501 1 200	Sampling Nb of defects	A R	125 2 3	125 3 4	125 5 6	125 7 8	125 10 11	4.3	5.4	7.4	9.4	12
				35 001 150 000	3 201 10 000	1 201 3 200	Sampling Nb of defects	A R	200 3 4	200 5 6	200 7 8	200 10 11	200 14 15	3.3	4.6	5.9	7.7	10
				150 001 500 000	10 001 35 000	3 201 10 000	Sampling Nb of defects	A R	315 5 8	315 7 8	315 10 11	315 14 15	315 21 22	2.9	3.7	4.9	6.4	9.0
				500 001 ∞	35 000 150 000	10 001 35 000	Sampling Nb of defects	A R	500 7 8	500 10 11	500 14 15	500 21 22	315 21 22	2.4	3.1	4.0	5.6	9.0
					150 001 500 000	35 001 150 000	Sampling Nb of defects	A R	800 10 11	800 14 15	800 21 22	500 21 22	315 21 22	1.9	2.5	3.5	5.6	9.0
					500 001 ∞	150 001 500 000	Sampling Nb of defects	A R	1 250 14 15	1 250 21 22	800 21 22	500 21 22	315 21 22	1.8	2.3	3.5	5.6	9.0
						500 001 ∞	Sampling Nb of defects	A R	2 000 21 22	1 250 21 22	800 21 22	500 21 22	315 21 22	1.4	2.3	3.5	5.6	9.0

Annex B (forming an integral part of this standard)

Example of the application of the table for a single plan

The inspection sheet defines the following parameters:
The level of inspection, the type of plan (single or double), AQL

Example | Level 3 | Single plan | AQL: 1

SINGLE PLAN

Number of parts			
Special levels of inspection			
S1	S2	S3	S4
2	2	2	2

	AQL (or ASL)					QL10 for AQL of:				
	0,65	1	1,5	2,5	4	0,65	1	1,5	2,5	4
Sampling	20	13	8	5	3					

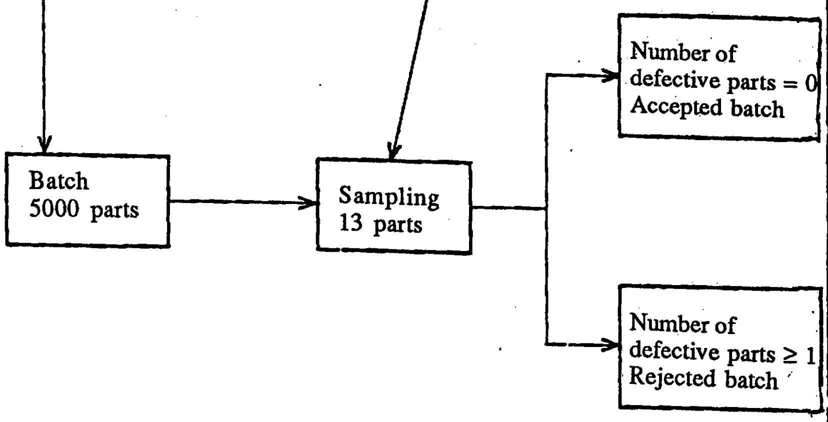
The inspector reads from the table: 5000 parts

the number of parts to be sampled
the decision to be taken - A: accepted
R: rejected

		3 201 35 000	501 1 200
		35 001 500 000	1 201 10 000

Sampling		20	13	32	20	20					
Nb of defects	A	0	0	1	1	2	11	16	12	18	25
	R	1	1	2	2	3					
Sampling		20	50	32	32	32					
Nb of defects	A	0	1	1	2	3	11	7,6	12	16	20
	R	1	2	2	3	4					

INSPECTION DIAGRAM



The client has less than 10% chance of accepting batches whose proportion of defective units is equal to or greater than 16%	LQ 16